Privacy-Preserving Disease Prediction using Federated Learning and Explainable AI

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***Abstract*—The healthcare industry is one of the most vulnerable to cybercrime and privacy violations because health data is very sensitive and spread out in many places. Recent confidentiality trends and a rising number of infringements in different sectors make it crucial to implement new methods that protect data privacy while maintaining accuracy and sustainability. Federated learning (FL) is a decentralized and privacy-protecting approach to deep learning and machine learning models. Advances in deep learning have transformed medical imaging, yet progress is hindered by data privacy regulations and fragmented datasets across institutions. In this article, we implement a scalable FL framework for interactive smart healthcare systems with intermittent clients using chest X-ray images. To address these challenges, we propose FedVGM, a privacy-preserving federated learning framework for multi-modal medical image analysis. FedVGM integrates four imaging modalities, including brain MRI, breast ultrasound, chest X-ray, and lung CT, across 14 diagnostic classes without centralizing patient data. Using transfer learning and an ensemble of VGG16 and EfficientNetB0 and, FedVGM achieved 96.88% testing accuracy, and 97.8% after fine-tuning. To ensure clinical interpretability, we apply Explainable AI techniques and validate results through performance metrics, statistical analysis, and k-fold cross-validation. FedVGM offers a robust, scalable solution for collaborative medical diagnostics, supporting clinical deployment while preserving data privacy.**

***Index Terms*—Federated Learning, Explainable AI, Data privacy, IEEE, Research, healthcare**

I. Introduction

MEDICAL imaging techniques such as magnetic resonance imaging (MRI), ultrasound, X-ray, and computed tomography (CT) play a critical role in modern healthcare, facilitating accurate disease diagnosis, treatment planning, and ongoing medical research. These modalities provide clinicians with detailed visualizations of internal structures, enabling the detection of anomalies at early stages, which is vital for improving patient outcomes. Among the most pressing healthcare concerns globally is cancer, a complex and life-threatening con- dition that continues to impact millions of people each year. According to the World Health Organization (WHO), the number of cancer cases is projected to nearly double within the next two decades, emphasizing the urgent need for scalable and efficient diagnostic tools . Brain, breast, lung, and chest-related cancers represent a significant portion of this global burden. For instance, brain cancer affects more than 250,000 people annually, often ac- companied by neurological impairments such as seizures and cognitive decline. Meningioma, the most common brain tumor, accounts for over 30%

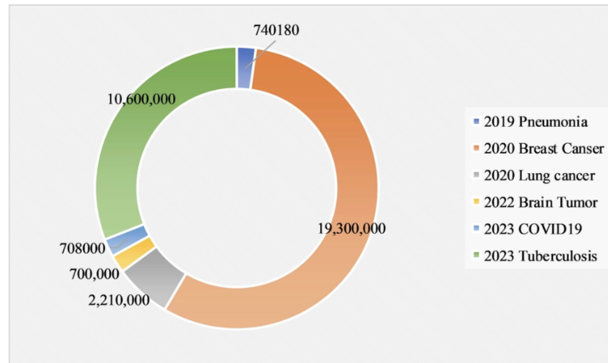

Fig. 1: Number of affected people with multiple diseases.

of such cases, with more than 700,000 Americans currently affected—a number that has risen sharply in the past three decades . Breast cancer is now the most frequently diagnosed cancer worldwide, with incidence nearly doubling since 2000 and reaching 19.3 million cases annually. Lung cancer, which remains difficult to detect in early stages, leads to over 2.21 million new cases per year, often associated with poor prognosis. While chest X-rays have long been a valuable diagnostic tool, their effectiveness is often limited by subtle imaging features and interpretive variability, which deep learning can help address. Fig. 1 shows the number of affected people with multiple diseases. Despite the potential of deep learning in medical imag- ing, challenges remain, such as limited access to multi-modal datasets and strict data privacy regulations, hindering the de- velopment of robust AI models. To address these challenges, federated learning (FL) has emerged as a privacy-preserving solution that enables collaborative model training across insti- tutions without exposing raw patient data as given by [24]. By keeping data local and sharing only encrypted model updates, FL ensures compliance with data protection laws while enabling institutions to benefit from collective learning. Importantly, FL can handle non-identical data distributions (non-IID) and heterogeneous imaging sources, making it suitable for complex, real-world clinical environments. However, clinical decision-making requires more than high performance. In practice, healthcare professionals must under- stand, verify, and trust AI outputs, particularly when

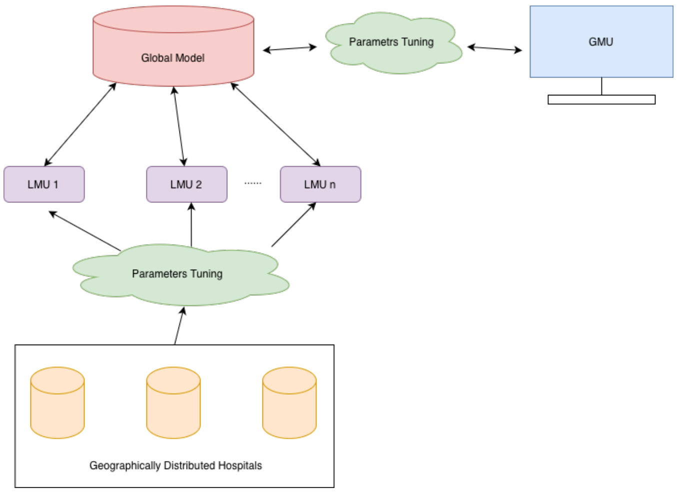

Fig. 2: Privacy preserving FL approach for decentralised healthcare informatics.

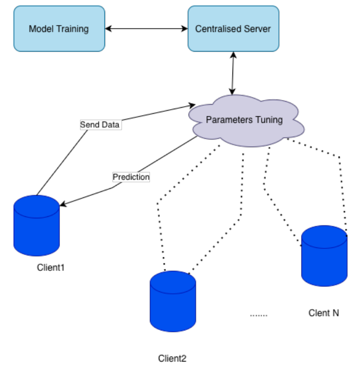

Fig. 3: Traditional model training with a client-server architecture.

diagnosing life-threatening conditions. Therefore, Explainable AI (XAI) [22] is essential in medical imaging, offering transparency and inter- pretability in model predictions. Techniques like SHAP (SHap- ley Additive exPlanations) and Grad-CAM (Gradient-weighted Class Activation Mapping) [23] visualize the most influential fea- tures or regions in an image, allowing clinicians to assess whether the model's reasoning aligns with medical knowledge. This interpretability helps avoid blind reliance on AI and builds trust in AI-assisted diagnostics. In this study, we introduce FedVGM, a federated learning framework for multi-modal medical image analysis. FedVGM integrates brain MRI, breast ultrasound, chest X-ray, and lung CT scans for 14 diagnostic classes without centralized data collection. Using transfer learning from ten CNN architectures, we propose an ensemble model combining VGG16 and MobileNetV2 to improve generalization. An ablation study evaluates the contributions of various model components, and we compare FL aggregation strategies, finding that median aggregation yields the best performance. The model is evaluated using multiple validation methods, and XAI techniques [25] are applied to improve interpretability. The major contributions of this work are as follows:

- FedVGM- A novel federated learning framework optimized for privacy-preserving multi-modal medical image analysis as sdepicted in Fig 2.
- A transfer learning–based ensemble model (VGG16 + EfficientNetB0) that achieves high classification performance across diverse datasets.
- A detailed evaluation of two FL aggregation (mean , median) strategies,
- Integration of explainable AI (SHAP, Grad-CAM) for enhancing transparency and clinical trust.
- Ablation studies that provide insights into model optimization and architecture refinement.
- The accuracy of 96.88% on the training dataset and 97.8% on fine tuning is achieved.
- A scalable, interpretable, and privacy - preserving framework suitable for real-world clinical deployment.

The rest of the paper is organized as follows: Section II on FL and medical image analysis. Section III describes the materials and methodology, including dataset details and the FedVGM architecture. Section IV presents experimental results and performance analyses. Finally, Section V concludes the paper by summarizing key contributions.

II. Related Work

Federated learning shows tremendous promise to enable privacy-preserving collaborative analytics on distributed medical images. This review analyzes pioneering techniques and accuracy outcomes on multi-modal use cases [30].

A. *Federated Learning in Medical Imaging*

Nazir et al. [2] provide a survey of FL applications across X-ray, CT, MRI, and histopathology, highlighting algorithms like FedAvg, FedSGD, SplitNN, and dynamic fusion for tasks such as classification and segmentation. Their findings show that FL often outperforms centralized learning while preserving data privacy. Similarly, Mahlool et al. [3] applied ranking weight aggregation to brain MRI datasets, achieving accuracy rates of 98% and 97.14%, outperforming traditional methods. Nguyen Tan et al. [6] used federated CNN and MobileNet models for breast cancer detection on the DDSM dataset, reaching 97.17% accuracy, emphasizing the importance of balanced aggregation.

In histopathology, Li L et al. [5] demonstrated FL's effectiveness for multi-site breast cancer classification with 91% accuracy. Moreover, Li Z et al. [7] introduced FedFocus for COVID-19 detection in chest X-rays, achieving 92% accuracy through weighted aggregation. Dou et al. [8] developed a cross-hospital FL framework for lung CT, obtaining an AUC of 95.27% and test precision of 88–96%, showing FL's clinical ap- plicability. Additionally, Haripriya et al. [9] integrated transfer learning with FL across three medical datasets, achieving high accuracy, but noting limitations such as reliance on non-medical pretraining and privacy risks, highlighting the need for further enhancement of FL scalability in real-world settings [26].

B. Privacy-Preserving and Secure Federated Learning

Data privacy is crucial in federated learning, especially in healthcare. Feki et al. [10] implemented a VGG16 and ResNet50-based FL framework on 108 COVID-19 chest X-rays, achieving 97% accuracy over 150 rounds while maintaining privacy. Similarly, Subashchandrabose et al. [11] proposed an ensemble FL framework for lung CT classification with 50 scans, incorporating differential privacy during optimization, achiev- ing 89.63% accuracy, surpassing traditional SVM and KNN classifiers. In genomics, Aziz et al. [12] introduced a privacy-preserving data sharing scheme with differentially private feature selection, achieving an AUC of 0.89 while maintaining a privacy budget. This demonstrates FL's potential in genomic research without compromising privacy.

C. Federated Learning Across Heterogeneous and Multi-Center Datasets

A key challenge in federated learning is managing heterogeneous data from diverse sources, which is common in multi-institutional medical environments. For instance, Naeem et al. [13] explored FL in brain tumor classification using CNNs, CapsNets, and U-Nets. Although their models performed well in metrics such as dice score and specificity, they faced issues like poor interpretability and inconsistent results due to dataset diversity. Similarly, Bhattacharya et al. [14] demonstrated that FL could maintain comparable AUC scores (88–93%) for COVID-19 detection using DenseNet121 on 3000 non-IID X-ray images. In addition, Hosseini et al. [15] proposed the Prop-FFL framework, which introduced fairness-aware optimization to mitigate bias across hospitals. Their approach using attention-gated multiple instance learning on histopathology datasets improved equity in performance, although limitations such as insufficient data from certain hospitals affected generalizability [29]. Moreover, Makkar and Santos [16] developed Se- cureFed, a robust FL framework that outperformed FedAvg and FedRAD, achieving 80%+ accuracy on 2100 X-rays, demonstrating fairness and resilience across varied client populations, though it lacked extension to multi-disease scenarios. Furthermore, Ullah et al. [17] focused on brain tumor segmentation using a U-Net-based FL framework, scaling from 50 to 100 clients, and yielding a dice coefficient of 0.89 and specificity of 0.96. On the other hand, Chowdhury et al. [18] trained multiple CNNs in a federated setup for COVID-19 classification and achieved 99.59% accuracy using Xception, though the study was constrained by the limited size of the training data. In contrast, Slazyk et al. [19] applied FL to pneumonia classification and segmentation on the RSNA dataset. Their findings revealed that while segmentation before prediction sometimes reduced accuracy, adding more clients and training rounds enhanced generalization [28].

D. Limitations of Previous Federated Models

Despite advancements in federated learning for medical imaging, several limitations remain. Many studies, such as Naeem et al. [13] and Mahlool et al. [4], report challenges with model interpretability and the lack of explainable AI, hindering clinical adoption. Similarly, Ullah et al. [17] and Feki et al. [10] demonstrate effective segmentation and classification but face scalability issues due to communication overhead and small-scale datasets. Furthermore, Subashchandrabose et al. [11] and Aziz et al. [12] highlight difficulties in generalizing across multi-site datasets, balancing privacy and model accuracy [27]. Additionally, issues like client drift, fairness disparities, and insufficient support for multi-disease tasks are identified in the works of Hosseini et al. [15] and Makkar and Santos. In summary, most previous studies focus on single-modality, single-disease tasks.

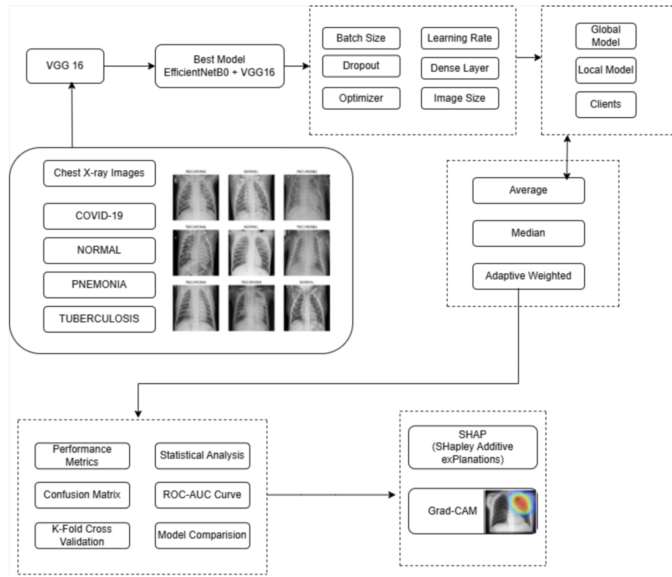

Fig. 4: Proposed Overview of Methodology

III. Methodology

This study utilizes chest X-ray images from four diagnostic classes, including pneumonia, COVID-19, and tuberculosis, obtained from the Chest X-ray (Pneumonia, COVID-19, Tuberculosis) dataset as shown in the Fig. 4. Standardize and purify the multi-class dataset before modeling. Transfer learning evaluates multiple CNN architectures pre-trained on natural images and fine-tuned with federated learning on medical images. Enhanced FedVGM uses EfficientNetB0 and VGG16 for above 96.88% testing accuracy, and 97.8% after fine-tuning . Ablation studies optimize FedVGM by removing model components to evaluate contributions. The federated training process consisted of multiple communication rounds between the server and clients. Every round select dynamic clients to participate in training. This Enhanced FedVGM uses Average, Median, and Adaptive Weighted aggregation for independent federated learning. The adaptive weighted method achieved the highest accuracy, topping 97.8% on the dataset. GradCAM verifies the reliability of the model.

The entire methodolgy is shown in the below Algorithm 1 and the flowchart illustrates the workflow of FedVGM 6.

Fig. 6 illustrates the workflow of the proposed FedVGM framework for privacy-preserving disease prediction. The pro-

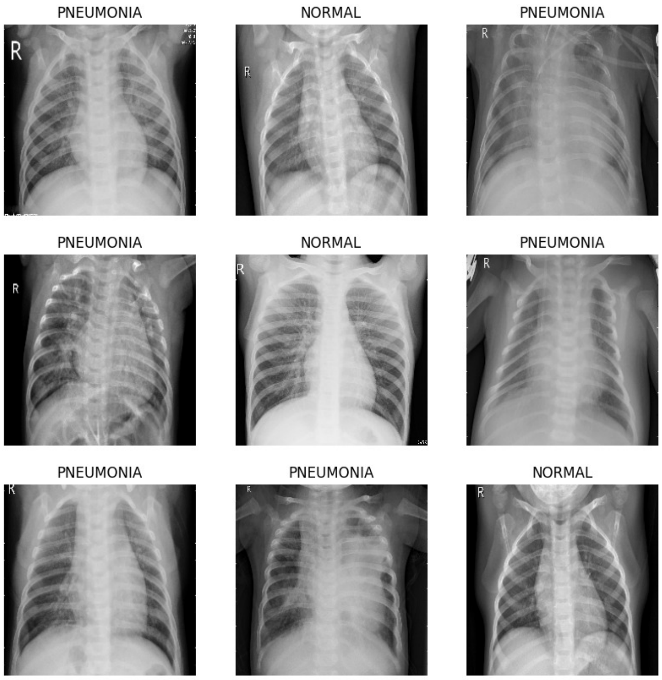

Fig. 5: Sample images from dataset

cess begins with the initialization of a global model G using pre-trained weights (VGG16/MobileNetV2). The system then enters in an iterative communication phase, where each round $t = 1$ to T involves the selection of a subset of participating clients C_t.

For each selected client $k \in C_t$, the global model is distributed, and local training is performed using the client's dataset D_k. This includes **fine-tuning** the model through transfer learning, followed by training for a fixed number of epochs E. After training process, each client computes its local model parameters W_k and sends them back to the central server.

At the server side, the received local weights are aggregated using a median-based aggregation strategy to obtain the updated global weights W_{global}. The global model is then updated accordingly and the process repeats for subsequent communication rounds.

Once the training rounds are completed, the final global model is evaluated on a test dataset. To enhance interpretability of the model, explainable AI techniques such as SHAP and Grad-CAM are applied to provide the model's predictions. The process concludes with the generation of the final, interpretable global model.

A. Data Description

Fig. 5 illustrates the class-wise distribution of the chest X-ray dataset with sample images from each category. The dataset consists of a total of 771 chest X-ray images categorized into four diagnostic classes: COVID-19 (106 images), Normal (234 images), Pneumonia (390 images), and Tuberculosis (41 images). These images are obtained from the Chest

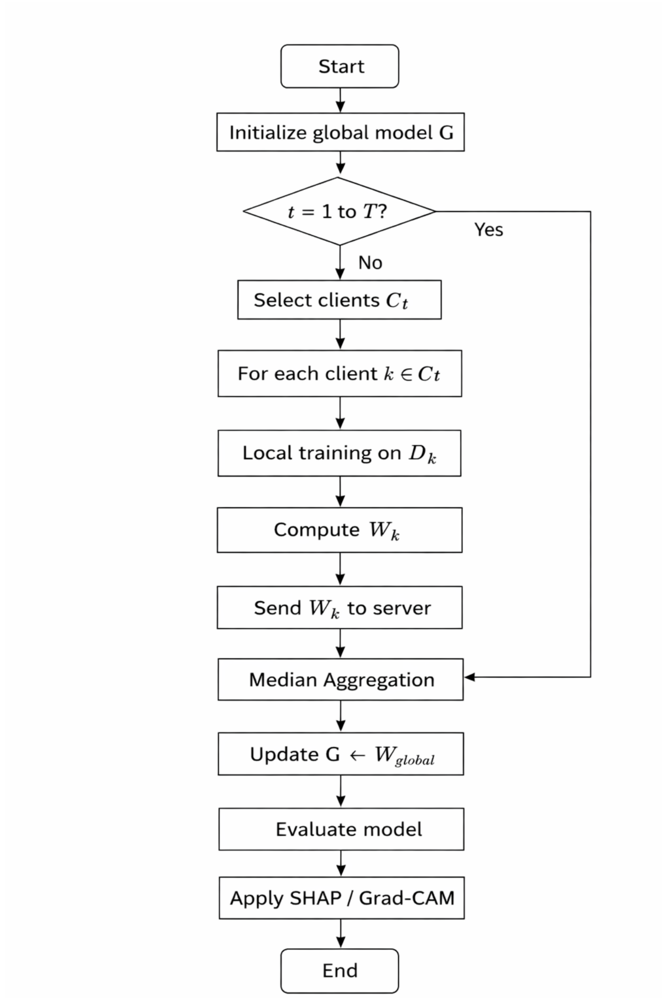

Fig. 6: Flowchart for the proposed methodology

Algorithm 1 FedVGM Framework

Initialize global model G with pre-trained weights (VGG16/MobileNetV2)
for each communication round $t = 1$ to T **do**
  Select subset of clients C_t
  for each client $k \in C_t$ **do**
    Receive global model G
    Load local dataset D_k
    Fine-tune model using transfer learning on D_k
    Train model for E epochs
    Compute local weights W_k
    Send W_k to server
  end for
  Aggregate weights:
  $W_{\text{global}} = \text{Median Aggregate}(W_1, W_2, \ldots, W_k)$
  Update global model $G \leftarrow W_{\text{global}}$
end for
Evaluate global model on test dataset
Apply SHAP / Grad-CAM for interpretability
return Final global model

X-ray (Pneumonia, COVID-19, Tuberculosis) dataset and are used for multi-class classification. The datasets were split into training and testing sets with an 80:20 ratio as depicted. The training set was used to optimize the deep learning models, while the test set was reserved for evaluating performance.

B. Model Description

The data was initially collected from Kaggle, specifically the Chest X-ray (Pneumonia, COVID-19, Tuberculosis) dataset, consisting of chest X-ray images categorized into four classes i.e. COVID-19, Normal, Pneumonia, and Tuberculosis classes. The data underwent standardization and cleansing followed by division for final modeling. Transfer learning was applied by using pre-trained deep CNN models on images, which were then refined by using federated learning on medical images. A multi-model solution based on the FedVGM framework was developed where EfficientNetB0 and VGG16 were used for improved feature extraction and classification performance. The framework aggregates client model updates using average, median, and different weighted aggregation strategies. The evaluation showed that the adaptive weighted aggregation performed overall best which leads to re-training of the final model. And after that GradCAM was used to explain model behaviour and to improve clinical interpretability. The framework was evaluated using performance metrics, confusion matrices, and accuracy comparisons. The goal was to create a precise, robust, and reliable architecture for clinical deployment.

C. Training Approach

In the federated learning framework, a global model is initialized at the central server and distributed to multiple local client nodes. Each client trains a local model using its own private dataset and updates the model parameters independently. These locally trained weights are then transmitted back to the server, where they are aggregated to update the global model. This iterative process continues across multiple communication rounds, improving the global model progressively.

To simulate real-world decentralized environments, dynamic client selection is employed, where only a subset of clients participates in each communication round. This enhances scalability and robustness, especially in non-IID data scenarios where client data distributions vary significantly.

The analysis used transfer learning models that were pre-trained on large-scale natural image datasets. The pre-trained weights were fine-tuned using federated learning across decentralized chest X-ray datasets. This enabled the models to adapt to the medical domain while preserving data privacy. The dataset was preprocessed using resizing and normalization to maintain consistency across all clients in the federated setup.

The aggregation of local model updates is performed using multiple strategies, including average, median, and adaptive weighted aggregation. The mathematical formulation for these aggregation strategies are as follows:

Average Aggregation:

$$\theta_{\text{new}} = \frac{1}{n} \sum_{i=1}^{n} \theta_i \tag{1}$$

Weighted Aggregation:

$$\theta_{\text{new}} = \sum_{i=1}^{n} w_i \theta_i \tag{2}$$

where θ_i represents the local model parameters from client i, and w_i denotes the weight assigned to each client, typically proportional to the client's data size.

Median Aggregation:

$$\theta_{\text{new}} = \text{median}(\theta_1, \theta_2, \ldots, \theta_n) \tag{3}$$

which provides robustness against outliers and noisy client updates.

Adaptive Weighted Aggregation:

$$\theta_{\text{new}} = \sum_{i=1}^{n} w_i \theta_i \tag{4}$$

$$w_i = \alpha \cdot \frac{n_i}{\sum n_i} + (1 - \alpha) \cdot Acc_i \tag{5}$$

Table I presents the performance of three aggregation strategies, namely Average, Weighted, and Adaptive methods, across multiple communication rounds. It can be observed that the performance varies across rounds due to the dynamic nature of client participation and data distribution.

The weighted aggregation method achieves superior performance in several rounds (e.g., Rounds 3, 6, and 9), indicating its effectiveness in assigning importance based on client contributions. The adaptive method also demonstrates competitive and stable performance, particularly in later rounds (e.g., Rounds 7 and 8), highlighting its ability to balance both data size and model accuracy.

In contrast, the average aggregation approach shows comparatively fluctuating performance, suggesting its limitation in handling data heterogeneity. Overall, the results indicate that weighted and adaptive aggregation strategies provide improved robustness and consistency compared to simple averaging in federated learning environments.

In our implementation, adaptive weighted aggregation is utilized, an addition to standard aggregation methods, an adaptive weighted aggregation strategy is proposed. The aggregation weights are dynamically adjusted based on both client data size and local model performance, improving overall model performance. This approach acts as a metadata-inspired enhancement, where clients with more representative data contribute more significantly to the global model as shown in Fig. 7 and Fig. 8..

The proposed model is an enhanced FedVGM framework that integrates EfficientNetB0 and VGG16 as parallel feature extractors. The classification heads are removed, and the extracted features are concatenated, followed by fully connected layers for multi-class classification across four categories.

TABLE I: Performance Comparison Across Communication Rounds

Round	Average	Weighted	Adaptive
1	0.9885	0.9365	0.9833
2	0.8500	0.8792	0.8896
3	0.9469	0.9781	0.9312
4	0.8583	0.8531	0.8375
5	0.9677	0.9417	0.9625
6	0.8938	0.9469	0.9125
7	0.8437	0.9312	0.9365
8	0.9156	0.9208	0.9677
9	0.9312	0.9833	0.9677
10	0.9260	0.8896	0.9104

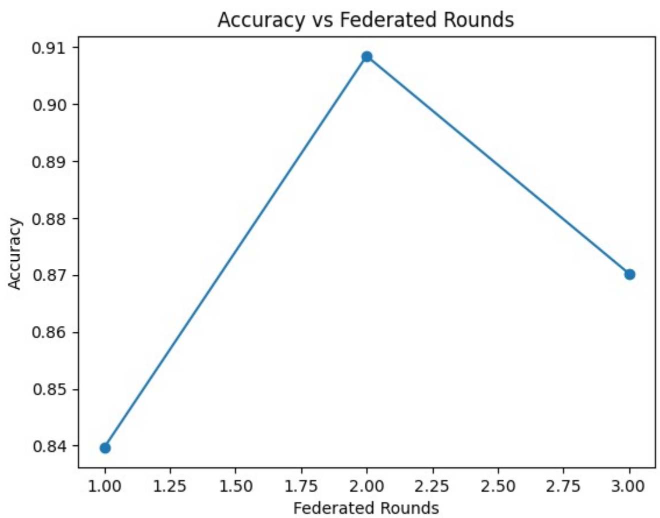

Fig. 7: Accuracy vs Federated Rounds in Static clients mode

An ablation study is conducted to evaluate the contribution of different architectural and training components, including learning rate, batch size, dropout, and aggregation strategies. The final model is selected based on its superior performance in terms of accuracy and robustness.

This federated learning setup ensures privacy-preserving training while effectively leveraging distributed medical data, leading to improved generalization and reliable clinical predictions.

The best-performing models were obtained by combining EfficientNetB0 and VGG16 through feature concatenation, followed by fully connected layers for multi-class classification across four categories. An ablation study was also conducted to systematically evaluate the contribution of different architectural and training components in the proposed Enhanced FedVGM framework.

D. Explainable AI

As AI systems grow more complex, the need for interpretability becomes challenging, especially in healthcare. It aims to make model decisions transparent and understandable. Two widely used XAI techniques are SHAP [20] and Grad-CAM [21]. SHAP quantifies how each input feature contributes to a prediction by measuring output changes when features are included or excluded. Grad-CAM, in contrast, visualizes the most influential image regions by computing gradients of the predicted class relative to convolutional feature maps by producing heatmaps of varying importance. Both methods increase transparency by identifying key features or regions involved in decision-making, which is essential for client trust. In our study, we integrated SHAP and Grad-CAM into the FedVGM to provide instance-level explanations. To assess their reliability, we compared highlighted regions with clinically relevant areas across tumor boundaries in MRI, lesion textures in ultrasound, and opacities in chest X-rays as shown in Fig. 9.

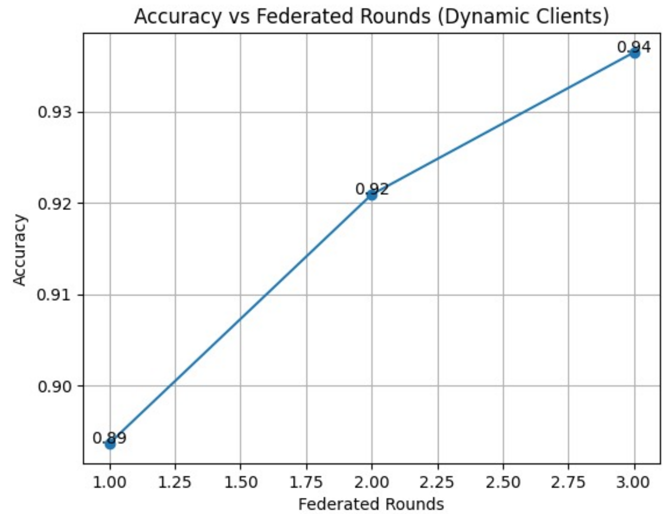

Fig. 8: Accuracy vs Federated Rounds in Dynamic clients mode

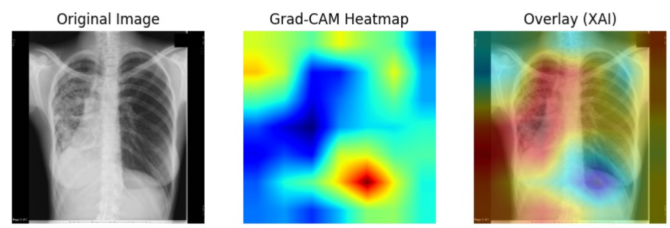

Fig. 9: Different images showing heatmap and XAI

IV. Results and Discussion

A. Performance Measures

We chose intermittent clients with varying amounts of test data to thoroughly evaluate the proposed approach. The proposed approach is tested on three publicly available standard datasets, the COVID, Pneumonia, Tuberculosis database images as shown in Fig 10, 11, 12 , 13.

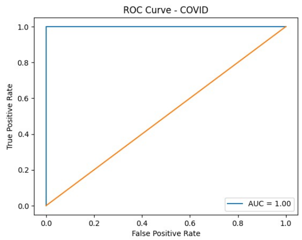

Fig. 10: ROC Curve - COVID

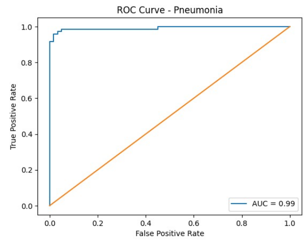

Fig. 11: ROC Curve - Pneumonia

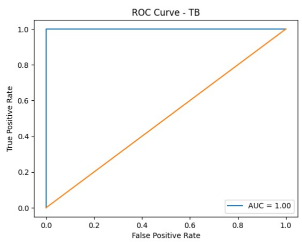

Fig. 12: ROC Curve - Tuberculosis

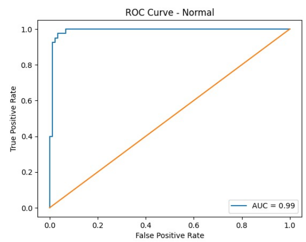

Fig. 13: ROC Curve - Normal

B. Evaluation Metrices

In this section, we outline the evaluation criteria used to assess the performance of the proposed FedVGM model. The performance metrics reported include training and testing accuracy, precision, recall, and F1 score. These metrics provide a comprehensive understanding of the model's performance across various aspects. The number of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) instances are used to calculate these matrices. The equations for these metrics are shown below : Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

F-measure:

$$\text{F-measure} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{8}$$

Kappa:

$$\kappa = \frac{\text{observedacc} - \text{expectedacc}}{1 - \text{expectedacc}} \tag{9}$$

where

$$\text{total} = TP + TN + FP + FN \tag{10}$$

$$\text{observedacc} = \frac{TP + TN}{\text{total}} \tag{11}$$

$$a = \frac{(TP + FN)(TP + FP)}{\text{total}} \tag{12}$$

$$b = \frac{(FP + TN)(FN + TN)}{\text{total}} \tag{13}$$

$$\text{expectedacc} = \frac{a + b}{\text{total}} \tag{14}$$

C. Result Analysis and Comparisons

Table II presents a comparison evaluation of different models, which includes MobileNetV2, VGG16, FedVGM, and the EfficientNetB0 + VGG16 ensemble. It is observed that the proposed EfficientNetB0 + VGG16 model achieves the highest performance among all evaluation metrics, with an accuracy of 0.9625, precision of 0.9688, recall of 0.9688, and F1-score of 0.9687.

MobileNetV2 and VGG16 give comparatively lower performance, indicating they have limited capability in capturing complex features from images. The proposed FedVGM framework shows a significant improvement over individual models by achieving an accuracy of 0.9312 thereby preserving the effectiveness of federated learning while preserving data privacy.

In short, the results show that after combining transfer learning with ensemble techniques and federated optimization, the model leads to much better performance and accuracy which makes the enhanced FedVGM approach suitable for real-world clinical applications and deployment.

TABLE II: Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1-score
MobileNetV2	0.8375	0.8642	0.8626	0.8623
VGG16	0.8562	0.9108	0.9062	0.9062
FedVGM	0.9312	0.9361	0.9313	0.9309
EfficientB0	0.9316	0.9567	0.9341	0.9700
EfficientB0 + MobileNetV2	0.8938	0.8790	0.9431	0.9654
EfficientB0+VGG16	**0.9625**	**0.9688**	**0.9688**	**0.9687**

D. Comparative Evaluation

In order to evaluate our model's accuracy, we compared our model to various other models using statistics like accuracy, precision, recall and F1 score. Our model outperforms all the other combinations of frameworks that we tested. It increases the accuracy to 96.88% and 97.8% with fine tuning. Using dynamic client participation instead of using a fixed set of clients in the training process models a real world federated scenario in which some clients may be available or unavailable for certain rounds. This confusion matrix shows the prediction and actual values comparison of the four classes that we have chosen as shown in Fig. 14

E. Analysis with explainable AI

SHAP plot analysis applied to all the datasets to improve interpretability. The SHAP results reveal strong positive diagonal signatures, indicating that the model accurately highlights discriminative features such as tumor textures, lung consolidations, or lesion shapes. These highlighted regions align well with clinical knowledge. In addition to high accuracy, SHAP-based transparency enhances trust of FedVGM across modalities.

Fig. 15 uses the Grad-CAM technique to visualize influential regions in the model's decision-making process [**?**].

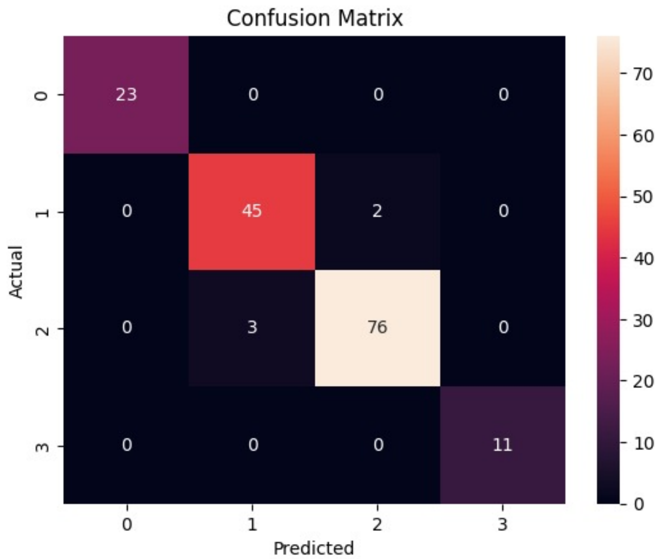

Fig. 14: Confusion Matrix for predicted and actual values of the four classes

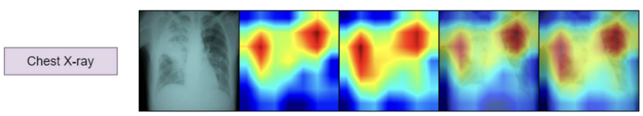

Fig. 15: GradCAM plots using FedVGM for Explainable AI

The color gradient red indicating strong relevance and blue indicating low or negative contribution which highlights areas critical to each prediction. Prominent red & yellow regions correspond to diagnostic features, while green & blue areas reflect background or less relevant features. This gradient-based explanation offers a detailed view of how spatial regions contribute to classification outcomes, enhancing model transparency.

To enhance clinical interpretability, SHAP and Grad-CAM outputs are generated after each prediction by the FedVGM model and presented alongside diagnostic results. SHAP highlights the contribution of each input region to the model's decision, while Grad-CAM visualizes key areas in the image that influenced classification. Notably, Grad-CAM provides more intuitive explanations which enables clearer identification of critical regions associated with different features.

V. Conclusion

This research proposes FedVGM, a novel federated learning framework optimized for multi-dataset medical image analysis. The model leverages transfer learning and ensemble techniques to improve accuracy within the federated loop. Evaluating median aggregation showed the best performance for consolidating client updates. The proposed approach was validated chest X-ray datasets, comprising 4 diagnostic categories with 771 images. FedVGM achieved exceptional accuracy of 96.88% on the aggregated dataset, and 97.8% after fine tuning. Extensive statistical and visual analysis using precision/recall metrics, ROC curves, confusion matrices verified model reliability. Integrated SHAP and Grad-CAM methods provided explainability by highlighting discriminative features and regions influencing predictions. This establishes trust and accountability for clinical adoption. FedVGM facilitates collaborative deep learning across decentralized medical systems, advancing cutting-edge privacy-preserving analytics. The optimized framework effectively consolidates distributed data to boost diagnosis while protecting sensitive information. In order to improve patient outcomes, this groundbreaking research extends federated learning capabilities to a variety of medical imaging modalities.

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